Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/17/2006
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	01/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	8
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date changed from 12/31/2006 to 9/30/2006 (1/2009) note - grant renewed for 4 more years/new grant number		
Key Personnel Changes/Previous PI:	2		
COI Name (Institution):	Chao, Beei-Huan (University of Hawaii) Sunderland, Peter (University of Maryland) Urban, David (NASA Glenn Research Center)		
Grant/Contract No.:	NCC3-1063/NCC3-1062		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed 'Flame Design' as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants. Recognizing the unique benefits of spherical diffusion flames, employ such flames to: 1. Obtain soot-inception limits as functions of stoichiometric mixture fraction, flow direction (direction across the flame front, i.e., towards the fuel or towards the oxidizer), amount of inert (nitrogen) and pressure for ethylene and methane flames. Identify the corresponding soot-inception temperature; 2. Determine effects of stoichiometric mixture fraction and flow direction on the phenomenon of permanently blue flames and, to this end, determine the effects of stoichiometric mixture fraction and flow direction on flame temperature, size, color and soot volume fraction; 3. Obtain extinction limits as functions of stoichiometric mixture fraction, flow direction, amount of inert and pressure for ethylene and methane flames. Identify the corresponding extinction temperature; 4. Evaluate whether this flame system can be used to obtain a pseudo-flammability limit for non-premixed flames in terms of stoichiometric mixture fraction and amount of inert. If so, determine these limits for ethylene and methane flames. Soot formation and flame extinction are problems of great concern in practical combustors. Recent normal-gravity experiments using counterflow diffusion flames have shown that permanently-blue flames (which are devoid of soot regardless of strain rate) and extinction-resistant flames can be produced for many hydrocarbon fuels by nitrogen exchange (increased stoichiometric mixture fraction). Furthermore, there is a unique stoichiometric mixture fraction wherein the flame temperature at extinction can be up to 200 K lower and the strain rate up to four times greater than that for the standard hydrocarbon/air flame with the same flame temperature. The present experiments employ microgravity conditions to allow for a unique set of experiments to clarify the underlying mechanisms responsible for flame response to stoichiometric mixture fraction.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As the goal of this program is to develop an understanding of combustion with the aim of reducing pollutants and increasing combustion efficiency, the results are directly applicable to terrestial power generation. In particular, with the present interest in carbon sequestration via oxygen-enriched combustion the present work is of particular importance as it identifies methods of controlling a number of pollutants through oxygen-enriched combustion.
Task Progress:	Testing in the 2.2-second drop tower has yielded sooting limits of spherical microgravity diffusion flames reacting ethylene and oxygen at various levels of dilution. The results provide a quantitative comparison of the effects on sooting limits of stoichiometric mixture fraction and convection direction. Testing in the drop tower also has identified a number of extinction limits. Normal-gravity tests involving coflowing gas jet flames have considered a similar range of stoichiometric mixture fraction for both normal and inverse conditions. These results lend support to the microgravity observations. Analytical modeling has advanced the understanding of the steady-state temperature, size, and residence times of the present spherical flames. Numerical modeling, which incorporates detailed chemistry and transport, has lent insight into the effects of radiation and unsteadiness of the present spherical flames. These results have been and are being disseminated in the publications listed in the publication section.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Sunderland, P.B., Urban, D.L., Stocker, D.P., Chao, B.-H. and Axelbaum, R.L. "Sooting Limits of Microgravity Spherical Diffusion Flames in Oxygen-Enriched Air and Diluted Fuel" Combustion Science and Technology, 176 2143-2164 , Dec-2004
Articles in Peer-reviewed Journals	Liu, S., Chao, Beei-Huan and Axelbaum, R.L. "A Theoretical Study on Soot Inception in Spherical Burner-Stabilized Diffusion Flames" Combustion and Flame 140 1-23 , Jan-2005
Papers from Meeting Proceedings	Z. Sun, R.L. Axelbaum, B.-H. Chao, P.B. Sunderland, D.L. Urban, and D.P. Stocker "Effects of Radiation Heat Loss and Lewis Number on the Temperature of Spherical Microgravity Diffusion Flames" Spring Technical Meeting of the Combustion Institute Central States Section, Austin, Proceedings, 5 pp. Mar-2004
Presentation	J.L. Taylor, P.B. Sunderland, Z. Sun, D.P. Stocker, D.L. Urban, B.-H. Chao, and R.L. Axelbaum "Images of Spherical Diffusion Flames" Art Exhibit Poster, 30th International Symposium on Combustion, Chicago (2004). Aug-2004
Presentation	J.L. Taylor, P.B. Sunderland, Z. Sun, D.P. Stocker, D.L. Urban, B.-H. Chao, and R.L. Axelbaum "Sooting Limits and Extinction of Spherical Diffusion Flames" Work-in-Progress Poster, 30th International Symposium on Combustion, Chicago (2004). Aug-2004